

Dual Nature of Radiation and Matter

Class: 12th

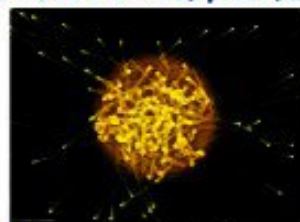
NCERT CHAPTER - 11th

Acc. to New Syllabus

Quantum Theory

PHOTONS: → Acc to Planck's quantum theory of radiation, an electromagnetic waves travels in the form of discrete packets of energy Called Quanta.

One quantum of light radiation is Called a PHOTON.



Characteristics of PHOTON

- 1) A photon travels with Speed of Light.
- 2) The Frequency of a photon does not change as it travels from One medium to another.
- 3) The Speed of photons changes as it travels through different media due to the change in its wavelength.
- 4) The Rest mass of a photon is zero - i.e a photon cannot exist at rest.

5) Energy of a photon, $E = h\nu = \frac{hc}{\lambda}$ h = plank's Constant λ = wavelength
 c = speed of light.

6) Momentum of a photon $p = mc = \frac{h\nu}{c} = \frac{h}{\lambda} = \frac{E}{c}$

7) From Einstein Mass-energy relationship, the equivalent mass m of a photon is given by, $E = mc^2 = h\nu$

OR

$$m = \frac{h\nu}{c^2} = \frac{E}{c^2}$$

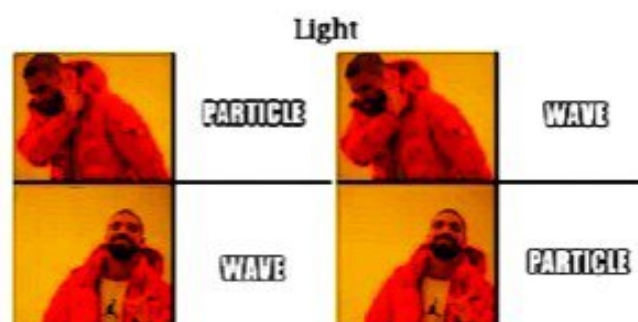
- 8) photons are electrically neutral and are not deflected by electric and magnetic fields.
- 9) In a photon-particle Collision, the total energy and total momentum are conserved. However, the Number of photons may not be conserved in a Collision. The photons may be absorbed or a new photon may be created.
- 10) photon Energy is independent of intensity of radiation.

PHOTOELECTRIC EFFECT

The phenomenon of emission of electrons from a metal surface, when electromagnetic radiations of sufficiently high frequency are incident on it, is called photoelectric effect.

Note: → The photo (light) - generated electrons are Called photoelectrons.

Note: → Alkali metals like Li, Na, K, Cs shows photoelectric effect with Visible light. Metals like Zn, Cd, Mg respond to U.V light.

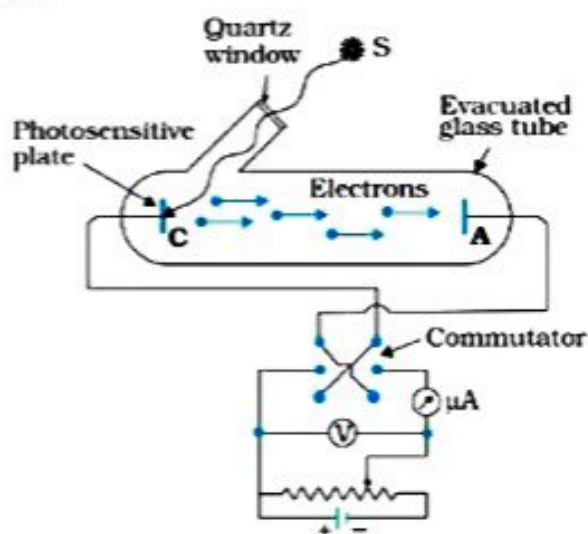


Experimental Study of PHOTO ELECTRIC EFFECT

To study photoelectric effect, an emitting electrode C of a photo sensitive material is kept at negative potential and a collecting electrode A is kept at positive potential in evacuated tube.

When a light of sufficiently high frequency falls on emitting electrode, photoelectron are emitted which travels directly to collecting electrode and hence an electric current called "photoelectric current" starts flowing in the circuit, which is directly proportional to the number of photoelectrons emitted by emitting electrode E.

When collecting electrode 'A' is made negative wrt emitting electrode C, an electric field is set up in the direction of motion of photoelectrons, which apply retarding force on electrons. So now lesser electrons reach the collecting electrode and photoelectric current in the circuit decreases.



Retarding Potential $\rightarrow$ is that potential at which no photoelectron reaches the collecting electrode A and the photoelectric current in the circuit becomes zero is called "Stopping potential".

Experiments on photoelectric effect

1) Effect of Intensity of light on photocurrent

i.e. Intensity of Photon $\propto$ Photocurrent (I_p)



Variation of photoelectric current with intensity of light.

$\rightarrow$ photoelectric effect is a One to One process.
i.e. One photon = One photoelectron

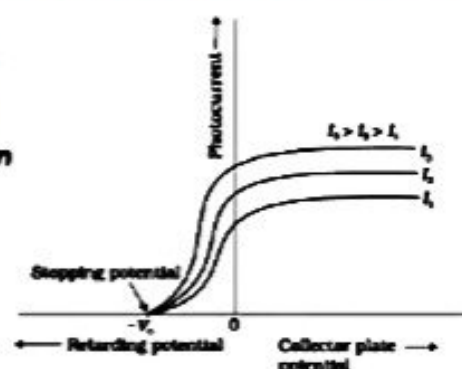
$\rightarrow$ $\uparrow$ intensity of photon $\rightarrow$ $\uparrow$ number of emitted electron (photoelectron)

2) Effect of potential on photoelectric current

i) At zero potential the emitted electron are in random motion, due to random motion of photoelectron a small current constitute i.e. At zero potential, photocurrent is not zero.

ii) When potential applied to collector plate is +ve. then most of the photoelectrons moves towards it and constitute photocurrent in the circuit i.e. photoelectric current increases with increase in accelerating voltage. At some stage, for a certain positive potential of plate A, all the emitted electrons are collected by the plate A and the photoelectric current becomes maximum or saturates.

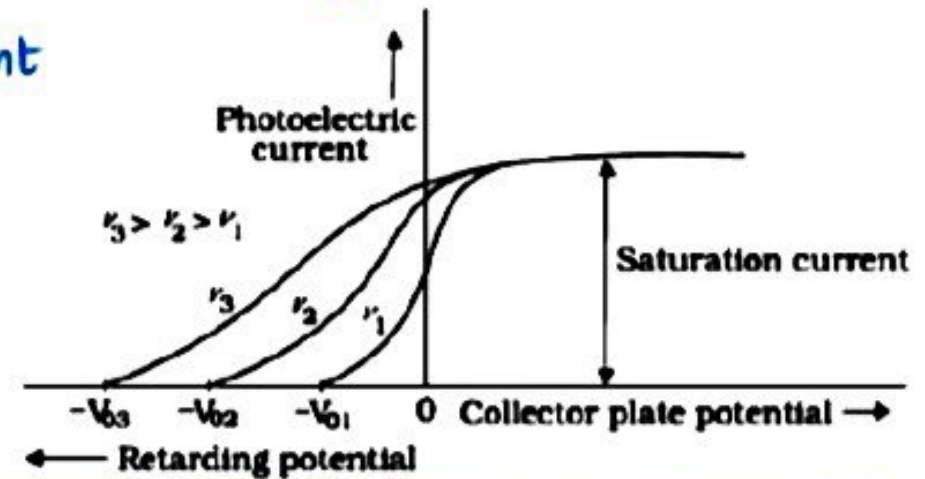
If we increase the accelerating potential of plate A further, the photocurrent does not increase. This maximum value of photocurrent is called **Saturation current**. Saturation current means when all the photoelectron reach the collector plate A.



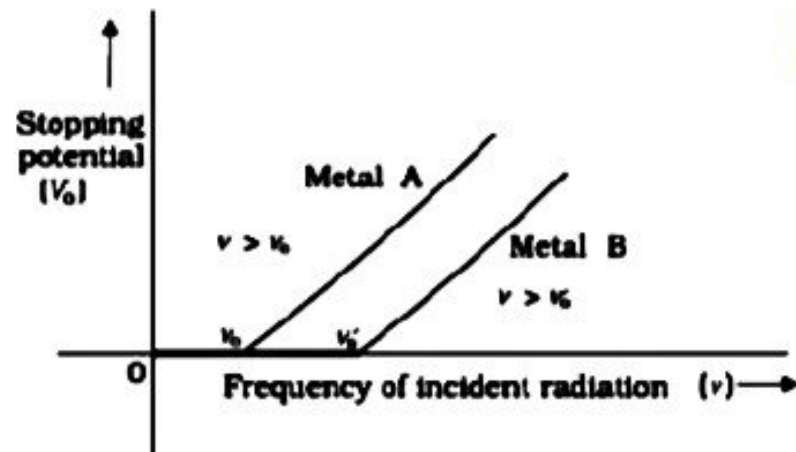
(iv) When Potential applied to Collector plate is Negative. the electrons are repelled and only the most energetic electrons are able to reach the plate A. At a certain value of Negative voltage V_0 on the plate A, photocurrent becomes zero, This minimum negative potential V_0 is called the "cut-off" or "Stopping potential".

3) Effect of Frequency of Incident radiation on Stopping Potential.

→ if radiation of same Intensity I but different frequencies ν_1 , ν_2 & ν_3 are incident on metallic surface, then it is found that photo electric current produced is same but the stopping potentials are different at different frequencies.



4) Variation of Stopping Potential V_0 with Frequency ν of Incident radiation for a given photosensitive Material.



Cut-off frequency $\rightarrow$ The minimum frequency ν_0 below which no emission takes place is called as "Cut-off frequency" or Threshold frequency.

Failure of Wave theory of Light to explain Photoelectric Effect

- 1) Acc. to wave theory, greater the intensity of radiation, greater are the amplitudes of electric and magnetic fields and hence greater is the energy density of wave. So the maximum kinetic energy of photoelectron emitted must depend on intensity of incident light, however practically it does not happen. So Independence of maximum kinetic energy of photoelectron emitted on intensity of incident light cannot be explained using wave theory of light.
- 2) Also, whatever the frequency of incident radiation may be, incident light of large intensity over a sufficient time must be able to impart enough energy to the electrons, so that they can get out of the metal surface, so a threshold frequency must not exist.
- 3) Further, number of electrons absorb energy continuously over the entire wavefront of the radiation. So energy absorbed per unit time by an electron becomes very small. So, in that case electrons may take quite long time to come out of metallic surface on continuous exposure of light on the surface. However, practically we found that there is no time lag between incident of light and emission of photoelectron.
So, we conclude that wave nature of light cannot be used to explain photoelectric effect.

PYQ: Discuss briefly how wave theory of light cannot explain photoelectric effect. (2/3 AI 2015)
Ans: In above Article.

Einstein's Explanation to Photoelectric effect (Previous Year's Question)

Einstein's Photoelectric Equation: $\rightarrow$ Acc. to Einstein, when light is incident on metal surface, incident photons are absorbed completely by valence electron of atoms of metal on its surface.

Energy $h\nu$ of each photon is partially utilised by an electron to become free or to overcome its "WORK FUNCTION" (W_0) and rest of the absorbed energy provides the maximum kinetic energy of photoelectron during the emission.

$$h\nu = \frac{1}{2}mv_{\max}^2 + W_0$$

$h\nu$ = Energy of photon
 W_0 = Work function.
 $\frac{1}{2}mv^2$ = K.E to the photoelectron.

Work Function (W_0) The minimum energy required by the valence electron of an atom on the surface of metal to become free or to become a photoelectron.

At threshold frequency ν_0 , the energy of photon $h\nu_0$ of incident radiation is just sufficient enough to liberate the electron i.e. just equal to work function W_0 .

$$W_0 = h\nu_0 = h\frac{c}{\lambda_0}$$

λ_0 = "Threshold frequency"

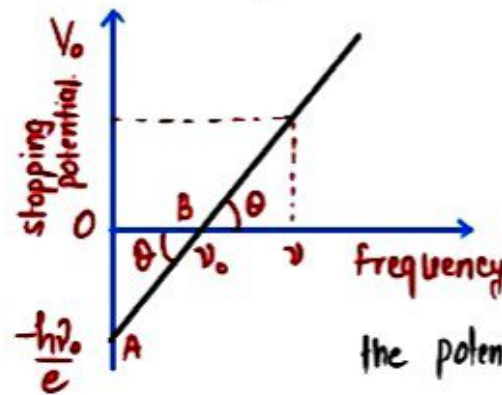
$$\text{So } h\nu = \frac{1}{2}mv_{\max}^2 + h\nu_0$$

$$\frac{1}{2}mv_{\max}^2 = h(\nu - \nu_0) \quad \text{This is called "Einstein's Photoelectric Equation".}$$

So at threshold frequency ν_0 , kinetic energy of photoelectron emitted is zero.

Explanation to Laws of Photoelectric Effect. (PYQ)

- (i). As soon as an electron absorbs energy of photon, it sets itself free. So emission of photoelectron starts as light falls on metal surface.
- (ii). Below threshold frequency ν_0 , energy of photon is less than work function of metal surface. i.e. less than minimum amount of energy required to liberate an electron. So emission of photoelectron take place only when the frequency of incident radiation is above or equal to the threshold frequency ν_0 . Further the slope of graph between stopping potential V_0 and frequency ν of incident radiation is



$$\tan \theta = \frac{V_0}{(\nu - \nu_0)} \Rightarrow \frac{eV_0}{e(\nu - \nu_0)}$$

$$= \frac{\frac{1}{2}mv_{\max}^2}{e(\nu - \nu_0)} = \frac{h(\nu - \nu_0)}{e(\nu - \nu_0)} = \frac{h}{e}$$

However the intercept of this curve on the potential axis is at $-OA = OB \tan \theta = \nu_0 \frac{h}{e}$

or $OA = -\frac{h\nu_0}{e}$

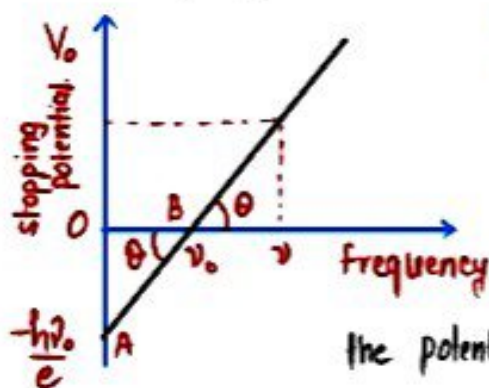
But if the graph is plotted between maximum kinetic energy $\frac{1}{2}mv_{\max}^2 = eV_0$ of photoelectron and frequency ν of incident radiation, then the slope of graph is equal to h , and intercept on the potential axis is at $-h\nu_0$.

- (iii). As energy of photon depends on the frequency of light, so the maximum K.E with which photoelectron is emitted depends only on the energy of photon or on the frequency of incident radiation.

- (iv). As the Number of photons in light depend on its intensity, and one photon liberates one photoelectron, so number of photoelectrons emitted depend only on the intensity of incident light.

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Wave Nature of Matter :

De-Broglie's Hypothesis :- (De-Broy) In 1924, French physicist Lewis de-Broglie suggested like radiation, matter too should have dual Nature. i.e. the particle like electrons, protons, neutron etc, can have particle as well as wave nature.

He reasoned that Nature was symmetrical and that two basic physical entities - matter and energy, must have symmetrical character.

He proposed that the wavelength λ associated with a particle of momentum 'p' is

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where m = mass of particle
 v = speed of particle.
 λ = wavelength of Matter wave



For a photon, as we have seen $p = \frac{h\nu}{c}$ therefore $\frac{h}{p} = \frac{c}{\nu} = \lambda$

→ λ is smaller for a heavier particle (large m) or more energetic particle. For example, the de-Broglie Hypothesis of a ball of mass 0.12 kg moving with a speed of 20 ms^{-1} is easily calculated.

$$p = mv = 0.12 \text{ kg} \times 20 \text{ ms}^{-1} = 2.40 \text{ kg ms}^{-1}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34} \text{ Js}}{2.40 \text{ kg ms}^{-1}} = 2.76 \times 10^{-34} \text{ m}, \text{ This wavelength is so small that is beyond any measurement.}$$

→ This is the reason why macroscopic objects in our daily life do not show wave-like properties.

De-Broglie Wavelength λ of the electron.

Consider an electron (mass m , charge e) accelerated from rest through a potential V . The kinetic energy K of the electron equals the work done (eV) on it by the electric field.

$$K = eV$$

Now $K = \frac{1}{2}mv^2$ we know $p = mv$
 so $v = \frac{p}{m}$

and $K = \frac{1}{2}m\frac{p^2}{m^2} \Rightarrow \boxed{K = \frac{p^2}{2m}}$ so that $p = \sqrt{2mK}$

$$\boxed{p = \sqrt{2meV}}$$

The de-Broglie wavelength λ of the electron is then

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

we get $\lambda = \frac{12.27}{\sqrt{V}} \text{ nm}$ substituting the numerical value of h, m, e
 V = magnitude of accelerating potential in volts.

PYQ: → A proton and an electron have same kinetic energy. which one has greater de-Broglie wavelength & why? (AI 2012)

Solution: → we know $\lambda = \frac{h}{\sqrt{2mK}}$ given $K_e = K_p$

$$\Rightarrow \lambda \propto \frac{1}{\sqrt{m}} \because m_p \gg m_e \therefore \boxed{\lambda_p \ll \lambda_e}$$

PYQ: → A proton and an electron have equal speeds. Find the ratio of de-Broglie wavelengths associated with them. (2020)

Solution: → $\lambda = \frac{h}{mv} \Rightarrow \frac{\lambda_p}{\lambda_e} = \frac{m_e}{m_p} = \frac{1}{1836}$ Answer.

Characteristics of matter waves

- (i) Matter waves are not electromagnetic waves in nature.
- (ii) Matter waves are non-mechanical waves. i.e. they can travel in vacuum.
- (iii) Matter waves are independent of charge. i.e. they are associated with every moving particle (whether charged or uncharged).
- (iv) Observation of matter waves is possible only when the de-Broglie wavelength is of the order of size of particle (i.e. the waves are diffracted).
- (v) The phase velocity of the matter waves can be greater than the speed of the light.
- (vi) The number of de-Broglie waves associated with n^{th} orbital electron is n .

Laws of Photoelectric Effect.

- (1) For a given material and a given frequency of incident radiation, the photoelectric current or number of photoelectrons ejected per second is directly proportional to the intensity of the incident light.
- (2) For a given material and frequency of incident radiation, saturation current is found to be proportional to the intensity of incident radiation, whereas the stopping potential is independent of its intensity.
- (3) For a given material, there exists a certain minimum frequency of the incident radiation below which no emission of photoelectron takes place. This frequency is called threshold frequency.
Above the threshold frequency, the maximum kinetic energy of the emitted photoelectrons or equivalent stopping potential is independent of the intensity of the incident light but depends upon only the frequency (or wavelength) of the incident light.
- (4) The photoelectric emission is an instantaneous process. The time lag between the incidence of radiation and emission of photoelectron is very small, less than even 10^{-9} s.